

AI2002 – Artificial Intelligence

Quiz – 3A

Name	
Roll. No.	
Section	

CLO3: Design basic AI-based solutions for real-world problems using appropriate techniques such as reasoning, tracking, or robot localization.

Scenario: Four tasks {T1, T2, T3, T4} must be assigned to shifts. Domains $D = \{\text{Morning (M), Afternoon (A), Night (N)}\}$. Constraints: $(T1 \neq T2)$, $(T2 \text{ before } T3)$, $(T3 \neq T4)$, $(T1 \text{ before } T4)$.
(Note: Before means Morning < Afternoon < Night.)

Part 1: Forward Checking (3 Marks)

Assume task T1 is assigned to the Afternoon (A) shift. Apply Forward Checking to the remaining tasks and update their domains in the table below.

Task	Initial Domain	After T1 = Afternoon (Forward Checking)
T2	{M, A, N}	
T3	{M, A, N}	
T4	{M, A, N}	

Part 2: MRV & Selection (4 Marks)

After Part 1, pick a task using the Minimum Remaining Values (MRV) heuristic. Assign it the earliest possible shift from its remaining domain, then apply Forward Checking again.

Selected Task: _____ Shift Assigned: _____

Remaining Task	Domain from Part 1	New Domain (After FC)

Does this branch result in a failure? [Yes / No]

Part 3: Constraint Inconsistency (3 Marks)

Shift assignment for Tasks {X, Y, Z} with Domains {Morning, Night}.

Constraints: (X before Y), (Y before Z), (X after Z).

1. If we assign X = Morning, what are the domains of Y and Z using Forward Checking?

$D(Y) = \{ \text{_____} \}$ $D(Z) = \{ \text{_____} \}$

2. Apply Arc Consistency specifically to the edge X after Z. Does it eliminate any values from X or Z?

Result: _____

3. Is this problem solvable? Briefly explain why based on your domain results.

Answer: _____